MIL-C-45135B(MU)
11 January 1963
SUPERSEDING
MIL-C-45135A(Ord)
27 May 1959

MILITARY SPECIFICATION

COMPUTER, BALLISTICS, M14

1. SCOPE

1.1 This specification covers one type of electromechanical computer known as Ballistics Computer, M14, of which the range servo amplifier is a separate component part. The computer transmits superelevation data to the fire control system of the 120 MM Tank M103A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts Equipment and Tools for Ordnance Materiel, Packaging of
MIL-S-16892	Synchros, 400 Cycle
MIL-I-45208	Inspection Requirements, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

F8602750	Computer, Ballistics, M14
LSQAP-8602750	List of Supplementary Quality Assurance Provisions for Computer, Ballistics, M14

PACKAGING DATA SHEET

MIL-P-14232/P8602750A	Packaging of Computer, Ballistics, M14 and M14E1

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 Materials.- The materials shall be in accordance with the drawings, material specifications, and general specification forming a part of this specification.

3.3 Design and construction.- The computer shall be manufactured in accordance with Munitions Command Drawing F8602750 Computer, Ballistics, M14 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure the inherent quality and performance capabilities.

3.4 General specifications.- The computer shall meet the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electron tubes
(f) Workmanship

3.5 Vibration.- The computer shall meet all requirements of the detailed specification, and shall not be damaged as a result of vibration in a vertical plane at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second for a duration of 5 minutes plus or minus 15 seconds.

3.6 Superelevation counter output accuracy.- With the computer having a constant load of between 90 and 110 ounce inches applied to the output shaft for periscope drive mount Drawing C8602819 the ammunition selector on the AP-T, M358 Cam, the range correction knob set to zero, and the mechanical calibration shaft set at 1500 yards, actuate the superelevation hand crank to match the outer index with the inner index within half line width by approaching the inner index from clockwise and counterclockwise directions without overtravel. Establish the position of the output shaft for the periscope drive mount to the 1500 yard check point by splitting the backlash error obtained from the clockwise and counterclockwise positions above. Superelevation counter accuracy shall be checked from the 1500 yard check point by determining if the rotations of the output shaft for the periscope drive mount is within the tolerances

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prescribed in Table I for the specified superelevation counter readings.

TABLE I

Superelevation counter output accuracy from the 1500 yard check point for the AP-T, M356 Cam (D8602528)

Position of output shaft for Drive Mount, M9	249°50' CCW	1500 yard starting point 0°	2R-326°10' CW	4R-110°10' CW
Counter reading in mils	3.0	6.47	21.0	28.0
Tolerance on output shaft position	±8°	±3°	±8°	±8°

3.7 Limits of shaft rotations.

3.7.1 Mechanical calibration shaft (Dwg. F8602555, Sheet 6).- The mechanical stops shall limit shaft travel within the range of 445-455 and 5175-5190 yards. The 455 yards index is 12R44°22' CW from the 1500 yard check point and the 5175 yard index is 24R275°24' CCW from the 455 yard index.

3.7.1.1 The electrical stops shall limit rotation of the mechanical calibration shaft between 480-500 yards and 5015-5080 yards and shall prevent the servo input motor from driving against the mechanical stop.

3.7.2 Output shafts.- The output shafts shall continually rotate during actuation of the hand crank from minus 1.5 plus or minus 0.1 mils to plus 68.5 plus or minus 0.5 mil as read on the superelevation counter.

3.8 Manual operation, input - output.- Using the AP-T, M358 Cam, the mechanical calibration shaft shall be positioned to each of the ranges as indicated in Table VI. Rotation of the hand crank in each direction to match the outer and inner indexes within plus or minus one half line width shall produce rotation of the output shaft for drive-mount to correspond to the values indicated.

3.8.1 The procedure of 3.8 shall be repeated for the 30 caliber BR-M1917 and the HE-T, M356 Cams.

3.9 Manual operation, range scale - output.- Using the AP-T, M358 Cam only, alinement of the outer index to the range scale by rotation of the hand

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crank in each direction shall produce rotation of the output shaft for drive-mount to correspond to the values of Table VI at ranges indicated therein.

3.10 Cam sequence.- Three cams shall have been placed in the computer in the following sequence starting with the AP-T, M358 followed by the BR-M1917 caliber 30 and the HE-T, M356 followed by three spacers. Cams used for inspection of the computer shall remain with that computer.

3.11 Stability.

3.11.1 Input motor.- The input motor shall not visually display self-sustained oscillatory operation at any range when the computer is energized by a range input signal at voltages and frequencies indicated below:

105 V. between 360 and 400 cycles A.C. and 18 V.D.C.
125 V. between 400 and 440 cycles A.C. and 28 V.D.C.

A maximum of three changes in direction of the input motor shall be considered as satisfactory operation. These three changes shall be counted as reversals in direction of the computer inner dial.

3.11.2 Output motor.- The output motor shall not visually display self-sustained oscillatory motion at the output shaft when the mechanical calibration shaft is rotated between minimum range and 1500 yards with the ammo-selector set on any cam. This shall be met with the computer energized with 28 plus or minus 0.5 volts dc, the output shaft to the periscope drive-mount loaded with 20 to 30 ounce inches and minus 15 percent range correction introduced. Each time the rotation of the calibration shaft is stopped, a maximum of eight changes in direction of the output motor shall be considered as satisfactory operation. These eight changes may be counted as reversals in direction of the computer outer dial.

3.11.2.1 Output servo sensitivity.- A load of 90 to 110 ounce inches shall be applied to the output shaft and the ammunition selector set on the AP-T, M358 Cam, the servo output motor energized with 18 plus or minus 0.5 volts dc, and the automatic follow-up cycle completed at any range between 4000 and 4800 yards. The number of times the contacts are required to close to advance the output shaft for periscope drive-mount, approximately 72° shall be determined by slowly rotating the mechanical calibration shaft in one direction. The average allowable superelevation increment for one make and break of the contacts shall not exceed 10°45' and the maximum shall not exceed 21°30'. Rotation of the mechanical calibration shaft in the opposite direction shall meet the above requirements.

3.12 Time of response.

3.12.1 Input motor.- The total elapsed time for the inner index to stabilize at the indicated range on the range scale after the input motor is energized shall not exceed the values in Table II at voltage indicated.

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TABLE II

Time of response-input motor

Range input signal transmitter	Inner index	Voltage ±0.5V DC	Elapsed time for inner index to stabilize (Seconds)
4800 yds.	500 yds.	28	3
500 yds.	4800 yds.	28	3
500 yds.	4800 yds.	18	4
4800 yds.	500 yds.	18	4

3.12.2 Output motor.- With a load of 90 to 110 ounce inches applied to the output shaft for the drive-mount, the total elapsed time for the outer index to stabilize in coincidence with the inner index, after the output motor is energized when the ammunition selector is set on HE-T, M356 Cam, shall not exceed the values in Table III at the voltage indicated.

TABLE III

Time for response-output motor

Position of outer index	Position of inner index	Voltage ±0.5V DC	Total elapsed time (Seconds)
4800 yds.	500 yds.	28	4
500 yds.	4800 yds.	28	4
500 yds.	4800 yds.	18	8
4800 yds.	500 yds.	18	8

3.13 Circuit breaker.

3.13.1 Input motor.

3.13.1.1 Motor stalled.- The circuit breaker shall open in 10 seconds or less when the mechanical calibration shaft is immobilized and a range signal is applied to the input synchro transmitter with the motor energized with 28 plus 0.5 volts.

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3.13.1.2 Continuous operation.- The circuit breaker shall remain in the closed position when the motor is energized with 28 plus 0.5 volts and subjected to twenty-five reversals over a period of 45 seconds.

3.13.2 Output motor.

3.13.2.1 Motor stalled.- With the output shaft immobilized, the outer and inner indexes separated by at least 200 yards, and the motor energized with 28 plus 0.5 volts, the circuit breaker shall open in 90 seconds or less.

NOTE: The above requirement shall be performed when the motor is "cold" approximately normal room temperature.

3.13.2.2 Continuous operation.- With a load of 90 to 110 ounce inches applied to the output shaft for the drive-mount the circuit breaker shall remain in the closed position when the mechanical calibration shaft is oscillated at a rate of four to four and one-half revolutions per second for 45 seconds over a spread of 1000 yards at ranges between 1200 and 2800 yards. This requirement shall be met with the ammo selector set on the AP-T, M358 Cam and the motor energized with 28 plus 0.5 volts dc.

3.14 Operability.- The operating torques shall not exceed the values of Table IV at temperatures indicated therein.

TABLE IV

Operating torque values

Control means	60° to 90° F, and 150°F	-40°F	Type of operation
Correction knob	34 oz. in.	45 oz. in.	Rotation
Correction knob	20 lb.	24 lb.	Pull out force
Ammo selector	40 lb. in.	80 lb. in.	Maximum lift of follower
Ammo selector	10 lb.	20 lb.	To translate follower
Hand crank	6 lb. in.	20 lb. in.	Rotate while engaged
Hand crank	12 lb.	20 lb.	To hold engaged
Circuit breakers	Electrically and mechanically operable	Electrically and mechanically operable	On - Off position
Interrupter Switch	"	"	In accordance with 3.20
Reset button	15 lb.	40 lb.	In accordance with 3.20

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3.15 Computer accuracy.

3.15.1 Electrical input.- The range signal from the input transmitter shall produce angular rotation of the mechanical calibration shaft from the 1500 yard index in accordance with the values in Table V.

TABLE V

Range input transmitter rotation as established by exact positioning of calibration shaft

Position of mechanical calibration shaft	11-R 68.8° CW ±1°	4-R 46.9° CW ±1°	*0° check point	2-R 334.6° CCW ±1°	7-R 21.6° CCW ±1°	9-R 356.5° CCW ±1°	11-R 305° CCW ±1°
Position of range input transmitter rotor	78°41' CCW	29°2' CCW	0°	20°36' CW	49°39' CW	70°15' CW	83°18' CW
Equivalent of range in yards	500	1000	1500	2000	3000	4000	4800

*Check point as established on mechanical calibration shaft in accordance with Drawing F8602555.

3.15.2 Output, electromechanical.- The requirements shall be met with the output motor energized with 28 plus or minus 0.5 volts then with 18 plus or minus 0.5 volts and a load of 90 to 110 ounce inches applied to shaft for periscope drive-mount.

3.15.2.1 Mechanical.- The position of the output shaft for the periscope drive-mount as determined from the position of the mechanical calibration shaft shall be in accordance with the values in Table VI.

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3.15.2.2 Electrical.- The position of the fine and coarse synchros as derived from the position of the mechanical calibration shaft shall be in accordance with the values in Table VII.

3.16 Backlash.- Backlash shall not exceed 1° at any setting at the output shafts to the periscope cant corrector, superelevation generator and the rangefinder cant corrector when shaft for periscope drive-mount is secured.

3.17 Environmental requirements.

3.17.1 Storage temperatures.- The computer shall be exposed to ambient temperatures of minus 80°F and plus 160°F until thermal equilibrium is reached. After thermal equilibrium has been re-established at room temperature, the computer shall show no evidence of physical failure such as casting failure or glass breakage, etc., and shall also meet the requirements of 3.18.

3.17.2 Operational temperatures.- The computer shall be exposed to ambient temperatures of minus 40°F and plus 150°F until thermal equilibrium is reached at each temperature. The computer shall then be operated with 18 plus or minus 0.5 volts, utilizing the HE-T, M356 Cam with a load of 90 to 110 ounce inches applied to drive-mount shaft. With range data introduced through a range input signal transmitter the computer shall be inspected at each temperature to the following requirements:

(a) Applicable requirements of 3.14.
(b) The outer index shall follow the inner index within one-half a graduation width when the inner index is advanced over its maximum excursion of the range scale in each direction.
(c) Requirements of 3.11 at 150°F only.

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TABLE VI

Position of Shaft for Periscope Drive-Mount, M9 as derived from position of mechanical calibration shaft
Based on 1500 yard reference range for AP-T, M358 Cam (Dwg. D8602528)

Range	500	1000	1500	2000	3000	4000	4800
Position of mechanical calibration shaft	11-R 68.8 CW	4-R 46.9° CW	0° check point	2-R 334.6° CCW	7-R 21.6° CCW	9-R 356.5° CCW	11-R 305° CCW
AP-T, M358 Drive-Mount, M9		162°43' CCW *4.21	0° *6.47	172°5' CW *8.86	1-R 183°36' *14.02	2-R 239°2' *19.79	3-R 249°7 *24.9
BR-M1917 Cal. 30 Drive-Mount, M9	135°22' CCW *4.59	1-R 227°31' CW *14.63	5-R 244°4' CW *34.86	----	----	----	----
Tolerance on output shaft to Drive-Mount, M9 for BR M1917 Cam Only	±10°48' *0.15	±18° *0.25	±39°36' *0.55	----	----	----	----
HE-T, M356 Drive-Mount, M9	176°24' CCW *4.02	130°19' CW *8.28	1-R 94°19' CW *12.78	2-R 75°12' CW *17.52	4-R 95°2' CW *27.79	6-R 204°29' CW *39.31	8-R 227°3 CW *49.4
Tolerance of Output shaft to Drive-Mount, M9 for M358 and M356 Cams	±10°48' *0.15	±10°48' *0.15	±7°12' *0.10	±10°48' *0.15	±18° *0.25	±28°48' *0.40	±39°3 *0.55

*Corresponding nominal Superelevation Value in mils (information only).

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TABLE VII

Position of rotation of rotors of the computer control transmitters as determined by position of external receivers at null position, based on 1500 yard reference range for AP-T, M358 Cam (D8602528)

Range		500	1000	1500	2000	3000	4000	4800
Position of mechanical calibration shaft		11-R 68.8° CW	4-R 46.9° CW	0° Check point	2-R 334.6° CCW	7-R 21.6° CCW	9-R 356.5° CCW	11-R 305° CCW
Cam	Synchro							
AP-T, M358	F		33°8' CW	0°	34°26' CCW	108°43' CCW	191°48' CCW	265°49' CCW
	*C		0°38' CW	0°	0°40' CCW	2°7' CCW	3°45' CCW	5°12' CCW
BR-M1917 Cal. 30	F	27°4' CW	117°30' CCW	1-R46°49' CCW	----	----	----	----
	*C	0°32' CW	2°18' CCW	7°59' CCW	----	----	----	----
Tolerance for BR-M1917 Cam Only		±2°9'	±3°35'	±7°53'				
HE-T, M356	F	35°17' CW	26°4' CCW	90°52' CCW	159°7' CCW	307° CCW	1-R 112°54' CCW	1-R 261°31' CCW
	*C	0°41' CC	0°31' CCW	1°46' CCW	3°6' CCW	6° CCW	9°12' CCW	12°8' CCW
Tolerance (Fine only) for M358 & M356 Cams	F	±2°9' *0.15 mil	±2°9' *0.15 mil	±1°26' *0.10 mil	±2°9' *0.15 mil	±3°35' *0.25 mil	±5°44' *0.40 mil	±7°53' *0.55 mil

NOTE: C = Coarse synchro F = Fine synchro * = Informational only

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3.17.2.1 Immediately upon completion of the environmental requirements of 3.17 the interior of the computer shall be dried by clean, dry, warm air of approximately 100° to 140°F.

3.17.3 Water resistance.- The interior of the computer shall show no evidence of water penetration after being subjected to a simulated rain fall as required by the rain test as specified in Specification MIL-F-13926, Procedure I. A time duration of fifteen minutes shall be substituted for two hours.

3.18 Post environmental check.- Computers that have been subjected to the environmental requirements of 3.17 shall upon return to room temperature (60° to 90°F) meet the requirements of 3.11, 3.15.1 and 3.15.2.2.

3.19 Illumination.- The internal lighting device of the computer shall provide adequate illumination to easily identify the inner and outer indexes of the range dial and the ammunition index when observed in a darkened area.

3.20 Interrupter switches.- When the computer hand crank is engaged, the interrupter switch shall open the motor circuit and energize the pilot light. Pressing of the reset button shall cause the output motor circuit to be closed and the pilot light to be extinguished. Actuation of the ammo selector handle shall cause the ammo selector interrupter switch to open the output motor circuit. Re-setting of the handle shall cause the switch to close the circuit.

3.21 The following electrical requirements may be determined any time prior to final acceptance of the computer.

3.21.1 High potential.- Insulation breakdown shall not occur between isolated circuits and between electrical circuits and body or frame when 500 volts (RMS) is applied for a period of between 5 seconds and 10 seconds.

3.21.2 Insulation resistance.- Following the high potential test, insulation resistance between ungrounded electrical circuits and the body or frame and between isolated circuits shall not be less than 5 megohms.

3.21.3 Continuity.- All electrical circuits shall be checked for continuity and shall conform to the wiring diagram.

3.21.4 Electrical zero.- Synchros shall be electrically zeroed in accordance with Drawings D8602859 and F8602555, Sheet 7.

3.22 Identification.- Each range servo amplifier shall be identified as a component of the computer with which it qualified by stenciling the serial number of the computer on the amplifier. Above two units shall then be packaged together.

3.23 Workmanship.- Workmanship shall be in accordance with workmanship requirements as specified in Specification MIL-F-13926.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements.- Inspection requirements and procedures shall be in accordance with Specification MIL-I-45208.

4.3 Initial production sample.- Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Initial production sample failure.- Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.3.2 Control sample.- Subsequent to the acceptance of the initial production sample 4.3, one unit selected at random, from each months production run, or one unit from each lot of 100, whichever occurs first (that have successfully passed the acceptance tests herein) shall be submitted to the Government for determination of compliance with 3.2 and 3.3. This inspection shall be conducted by the contractor, with surveillance by the Government at the contractor's facility or Government approved laboratory, using the equipment authorized on the contract.

4.3.3 Control sample failure.- Should a control sample fail to pass the surveillance tests, the contractor shall stop all operations governing the manufacture of components or assemblies rejected for material or construction requirements not in accordance with mandatory drawings and specifications. The contractor shall make the necessary corrections and re-submit new control samples. Acceptance of the corrected units shall be delayed until the new control samples have been approved.

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4.4 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.1 Defective units and lots.- Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.5 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection of components.- Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP)-8602750.

4.6.1 Contents of SQAP.- The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of each component covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual nonconforming items.

4.6.2 Contractor's inspection for all components not covered by SQAP.- Inspection for these components shall be in accordance with the inspection procedures of the inspection system required by Specification MIL-I-45208. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the ballistics computer.

4.7 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.7.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance except as specified in 4.7.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7.2 Inspection equipment design requirements.

4.7.2.1 Device to test vibrating requirements.- The special test equipment shall consist of a vibrating machine which will produce vertical motion at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion), and shall also be equipped with a medium capable of holding the computer assembly securely to the vibrating machine. The medium shall hold the computer assembly in the normal operating position with

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periscope drive-mount output shaft clamped in a secure position. (Two clamping means may be required for different locations of the output shaft.) The vibrating equipment must meet the tolerances specified in Specification MIL-F-13926.

4.7.2.2 Equipment to test functional requirements.- Final inspection test equipment shall be capable of determining acceptable or rejectable computers for those requirements specified in 3.6 to 3.21 inclusive. The special test equipment shall be so constructed as to permit location of the Ballistics Computer, M14 in its normal operating position, and shall have sufficient mediums to test the requirements as listed above. The mediums noted below shall be to the accuracies stated. The medium used to view the inner and outer index shall be 3 power or better with a line width of which shall be 0.008 plus or minus 0.001. The medium used to view the superelevation counter of the ballistics computer shall be 3 power or better. The medium used to supply power for testing the requirements of stability, time of response, circuit breaker, computer accuracy and environmental requirements shall be as follows:

(a) Input voltage range - 95 to 130 volts a-c maximum - input current - approximately 35 amperes a-c.
(b) Output voltage range - 3 to 30 volts d-c continuously adjustable.
(c) Regulation accuracy - plus or minus 0.25% from 95 to 130 volts a-c - line change at any fixed load from 3 to 30 amperes d-c.
(1) Plus or minus 2% from 95 to 130 volts a-c - line change at any fixed load from 0 to 3 amperes d-c.
(2) Plus or minus 0.25% from 3 to 30 amperes d-c - load change at any fixed line voltage from 95 to 130 volts a-c.
(3) Plus or minus 2% from 0 to 30 amperes d-c - load change at any fixed line voltage from 95 to 130 volts a-c.
(d) Recovery time - 0.2 seconds
(e) Input frequency range - 50 to 60 cycles per second.
(f) Load range - 0 to 30 amperes.
(g) Ripple - 1% RMS maximum of the output setting - this rating applies to a nominal input frequency of 60 cycles at normal room temperature.
(h) This unit will operate at ambients from minus 40°F to plus 150°F.

The medium used to establish pertinent positions of the mechanical calibration shaft while testing the requirements of: superelevation counter output accuracy, mechanical calibration shaft, electrical stops, manual operation input - output, manual operation, range scale - output, output motor, output servo sensitivity, input motor - motor stalled, output motor - motor stalled, continuous operation, computer accuracy - electrical input, computer accuracy - mechanical, and computer accuracy - electrical, shall be graduated in both yards and degrees with an angular tolerance of plus or minus 5 minutes total. The medium shall incorporate a means of locking. There shall also be a method to establish amounts of rotation. The medium used to establish pertinent positions of the output shaft periscope drive mount while testing the requirements of superelevation

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counter output accuracy, manual operation input - output, manual operation, range scale - output, output motor, output servo sensitivity, output motor - motor stalled, continuous operations, computer accuracy - output electromechanical, computer accuracy - mechanical, and operational temperatures, shall be graduated in degrees with an angular tolerance of plus or minus 9 minutes total. The medium shall incorporate a means of locking. There shall also be a method to establish amounts of rotation. The medium shall incorporate a method of torque, adjustable, within 20 - 110 ounce inches. The medium used to check backlash of output shafts to the persicope cant corrector, superelevation generator and the range finder cant corrector shall be graduated 360 degrees in 1 degree increments with an angular total tolerance to be within plus or minus 6 minutes. The medium used to establish pertinent positions of the range signal while testing the requirements of: stability - input motor, time of response - input motor, time of response - output motor, circuit breaker - continuous operation, computer accuracy - electrical input, and operational temperatures, shall be graduated from 440 yards to 5190 yards within total angular tolerance plus 0.5 minutes, and shall be equipped with a method of transmitting this range electrically to the ballistics computer through amplifier. The medium used to establish pertinent positions of the superelevation output signal receiver while testing the requirements of computer accuracy - electrical - coarse and fine synchros, shall be graduated a full 360 degrees to the 1/2 degree with a total tolerance plus or minus 1 minute and shall be equipped with a method of receiving the electrical output from the ballistics computer.

4.7.2.3 Device to test torque requirements - correction knob - rotation.- The device shall have a minimum range of from zero to 45 inch ounces, the accuracy of which shall be plus or minus 1/4 inch ounces throughout the entire zone of excursion. The device shall be equipped with adapter to accommodate testing of the correction knob in accordance with Table IV.

4.7.2.4 Device to test force requirements - correction knob - pull.- This equipment shall have a minimum range of from zero to 24 pounds, the accuracy of which shall be plus or minus 1/4 pound throughout the entire zone of excursion, the device shall be equipped with adapter to accommodate testing of the correction knob.

4.7.2.5 Device to test torque requirements - ammo selector - rotation.- The special test equipment shall have a minimum range of from zero to 80 pound inches throughout the entire zone of excursion. The device shall be equipped with adapter to accommodate testing of the ammo selector.

4.7.2.6 Device to test force requirements - ammo selector - pull.- This special test equipment shall have a minimum range of from zero to 20 pounds the accuracy of which shall be plus or minus 1/4 pound throughout the entire zone of excursion. The device shall be equipped with adapter to accommodate testing of the ammo selector.

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4.7.2.7 Device to test torque requirements - hand crank - rotation.- The special test equipment shall have a minimum range of from zero to 20 pound inches, the accuracy of which shall be plus or minus 1/4 pound inch throughout the entire zone of excursion. The device shall be equipped with adapter to accommodate testing of the hand crank.

4.7.2.8 Device to test force requirements - hand crank - push.- The special test equipment shall have a minimum range of from zero to 20 pounds, the accuracy of which shall be plus or minus 1/4 pound throughout the entire zone of excursion. The device shall be equipped with adapter to accommodate testing of the hand crank.

4.8 Acceptance inspection.

4.8.1 Classification of defects.- The classification of defects in Tables VIII and IX shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The AQL specified pertains to each individual characteristic.

4.8.2 Samples selected.- Samples selected for acceptance inspection listed under Table VIII shall include those samples used for acceptance inspection specified in Table IX.

4.8.2.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be re-submitted for acceptance as outlined in Standard MIL-STD-105.

TABLE VIII - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, in Table III-B with Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0 percent Defective	Req't. Par.	QA. Par.	Test Method
101. Storage temperature	3.17.1	4.9.1	Visual
102. Operating temperature	3.17.2	4.9.2	Special Test Equipment (STE)
103. Vibration	3.5	4.9.3	STE
104. Improper sealing	3.17.3	4.9.24	Visual
105. Physical failure - Post environmental	3.18	4.9.25	Visual

MINOR: NONE DEFINED

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TABLE IX - CLASSIFICATION OF DEFECTS

Use Inspection Level II, in Table III-A with Sampling Plan IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5 percent Defective		Req't. Par.	QA. Par.	Test Method
101.	Insufficient superelevation counter output accuracy	3.6	4.9.4	Special Test Equipment (STE)
102.	Mechanical stops, not within limit	3.7.1	4.9.5	STE
103.	Electrical stops, not within limit	3.7.7	4.9.5.1	STE
104.	Output shafts, not within limits	3.7.2	4.9.6	STE
105.	Manual operation, input - output, not within limits	3.8.1	4.9.7	STE
106.	Manual operation, range scale, output, not within limits	3.9	4.9.8	STE
107.	Cam sequence, operation	3.10	4.9.9	Visual
108.	Failure, input motor, not within limits	3.11.1	4.9.10	STE & Visual
109.	Output motor, not within limits	3.11.2	4.9.11	STE & Visual
110.	Output servo sensitivity, not within limits	3.11.2.1	4.9.12	STE
111.	Input motor, time response, not within limits	3.12.1	4.9.13	STE & SME (Standard Measuring Equipment)
112.	Output motor, not within limits, (time response)	3.12.2	4.9.14	STE & SME
113.	Motor stalled, not within limits, (input)	3.13.1.1	4.9.15	STE
114.	Continuous operation (failure),	3.13.1.2	4.9.16	STE
115.	Motor stalled, not within limits, (output)	3.13.2.1	4.9.17	STE
116.	Continuous operation, failure	3.13.2.2	4.9.18	STE
117.	Operability, excess torque	3.14	4.9.19	STE
118.	Electrical input, not within limits	3.15.1	4.9.20	STE
119.	Mechanical output, not within limits	3.15.2	4.9.21	STE
120.	Electrical output, not within limits	3.15.2.2	4.9.22	STE
121.	Backlash, excess	3.16	4.9.23	STE
122.	Insufficient - illumination	3.19	4.9.26	Visual

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TABLE IX - CLASSIFICATION OF DEFECTS cont'd

Use Inspection Level II, in Table III-A with Sampling Plan IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5 percent Defective	Req't. Par.	QA. Par.	Test Method
123. Inoperable interrupter switches	3.20	4.9.27	Visual
124. Insulation breakdown, high potential	3.21.1	4.9.28	SME
125. Insufficient insulation resistance	3.21.2	4.9.28.1	SME
126. Lack of continuity	3.21.3	4.9.28.2	SME
127. Improper electrical zero	3.21.4	4.9.28.3	STE
128. Improper identification	3.22	4.9.30	Visual

MINOR: NONE DEFINED

4.9 Test methods and procedures.

4.9.1 Storage temperature.- This test shall be conducted in accordance with temperature cycling Procedure II, condition "A" and "H" in Table I of Specification MIL-F-13926. Upon completion of the specified exposure in 3.5, the computer shall be inspected for any physical damage, and then, subjected to all other tests specified in this specification. These tests shall be performed on the initial production sample only.

4.9.2 Operating temperature.- This test shall be conducted in accordance with Procedure I, of Specification MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" and or raised beyond condition "G". When the computer is at the indicated temperature it shall be subjected to the tests specified in 3.6. The controls of the computer shall meet the specified torque, force and applicable requirements of 3.20 for the indicated operating temperature and shall employ special test equipment for complete inspection.

4.9.3 Vibration.- The computer shall be mounted on the vibrating equipment in its normal operating position. The periscope drive mount output shaft shall be supported and immobilized and shall remain immobile throughout the test and vibrated in accordance with 3.5.

4.9.4 Superelevation counter output accuracy.- Superelevation counter output accuracy shall be inspected to the requirements of 3.8 and Table I when mounted on special test equipment 4.7.2.2 and applying a constant load as required on the periscope drive mount output shaft. A visual reading of the range scale and the inner matching dial index shall be made. The superelevation counter scale shall be read with an accessory of the special test equipment, having a reticle line for reading to (0.01) mil accuracy. The mechanical

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calibration shaft shall be inspected for input yardage and revolutions.

4.9.5 Mechanical calibration shaft.- This test shall be accomplished with special testing, as outlined in 4.9.4 and it shall meet the requirements of 3.7.1. The shaft shall be rotated slowly when approaching stops. The stops should not be bumped with force.

4.9.5.1 The electrical stops shall be tested with special testing equipment capable of reading the range and electrically indicating when the servo motor is not energized to meet the requirements of 3.7.1.1.

4.9.6 Output shafts.- Rotation of the output shafts shall be visually observed during actuation of the hand crank. The special testing equipment in 4.7.2.2 shall be used to meet the tolerance requirements of 3.7.2.

4.9.7 Manual operation, input - output.- With the mechanical calibrated shaft affixed with special testing equipment as outlined in 4.7.2.2 and the output shaft for the drive mount affixed with special testing equipment capable of measuring angular rotation in revolutions, rotation of the hand crank, in each direction to match the outer and inner indexes, when viewed through the special testing equipment. Equipment for the test shall meet the manual operation requirements of 3.8 and 3.8.1.

4.9.8 Manual operation, range scale - output.- Inspection shall be conducted using special testing equipment to read the range scale and a device to measure rotation of the output drive mount shaft. Rotation of the hand crank in each direction shall be conducted to aline the outer index to the range scale. This tests shall be conducted to determine compliance with 3.9.

4.9.9 Cam sequence.- Shall be visually inspected to insure that the sequence is correct as required in 3.10. The cams used shall be the same with which the computer qualified.

4.9.10 Input motor.- Special testing equipment shall be used capable of supplying an electrical range signal input to the computer, and a dc power supply capable of such capacity and stability as to maintain the specified voltages within three percent during operation of the computer. The ac power supply shall be capable of maintaining the specified voltages and frequencies within ten percent during operation of the computer. A check for oscillatory operation and the permissible maximum of three changes in direction of the input motor can be visually observed on the computer inner dial to meet the requirements of 3.12.1.

4.9.11 Output motor.- With the constant specified load applied to the periscope drive mount output shaft and the range input indicated on the mechanical calibrated shaft, a check for oscillatory operation and the permissible maximum of eight changes in direction of the output motor can be visually observed on the computer outer range dial to meet the requirements of 3.12.2.

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4.9.12 Output servo sensitivity.- The periscope output drive mount shaft shall have the specified constant load applied and, the range shall be read from the range scale. Inspection shall be made with special test equipment. Angular rotation of the periscope drive mount shaft shall be indicated as specified when the mechanical calibrated input shaft is slowly rotated. The servo clicks shall be counted and averaged out to meet the requirements of 3.11.2.1.

4.9.13 Input motor.- With the servo input motor circuit breaker in the off position, the special test equipment shall be capable of electrically transmitting a range signal input at the specified ranges in Table II. The inner index shall be positioned at the range specified. The elapsed time (after the power has been applied by re-setting the circuit breakers) for the inner index to stabilize within half a line width on the range scale, when timed with standard measuring equipment, shall not exceed the tolerance requirements specified.

4.9.14 Output motor.- With the specified constant load applied to the periscope drive mount shaft using special test equipment and the inner index positioned to the specified ranges, the hand crank shall be rotated to the specified range for the outer index. The elapsed time after the power has been applied, by re-setting the circuit breaker for the inner index to stabilize within half a line width on the range scale, when timed with standard measuring equipment, shall meet the requirements of 3.12.2 and Table III.

4.9.15 Motor stalled.- With the mechanical calibration shaft locked by special testing equipment and the motor energized at the specified voltage, the length of time for the circuit breaker to open, shall be timed with standard measuring equipment, to meet the requirements of 3.13.1.1.

4.9.16 Continuous operation.- Special test equipment that is capable of electrically transmitting range to the computer shall be used for subjecting the motor to the required reversals. Standard measuring equipment shall be used to time the continuous operation as specified in 3.13.1.2.

4.9.17 Motor stalled.- With the periscope drive mount output shaft locked by the special test equipment and the motor energized at the specified voltage, the length of time for the circuit breaker to open shall be timed with standard measuring equipment, to meet the requirements of 3.13.2.1.

4.9.18 Continuous operation.- The specified load on the output shaft for the periscope drive mount shall be applied by the special testing equipment. The mechanical calibrated input shaft shall be oscillated at the specified rate, over the required spread of ranges by use of standard measuring equipment to meet the requirements of 3.13.2.2.

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4.9.19 Operability.- The torques and forces required to actuate the controls of the computer shall not exceed the applicable values specified in Table IV and 3.20 while at the ambient temperature of plus 60°F to plus 90°F. These tests shall be conducted with special test equipment and standard measuring equipment capable of measuring the required torques and forces in 3.14.

4.9.20 Electrical input.- This inspection test shall be conducted utilizing the special test equipment used in 4.13. The range input signal shall be used for checking the angular rotation of the mechanical calibration input shaft specified in Table V. With the periscope drive mount output shaft having the constant specified load applied, and the output motor energized with the required voltages, the following tests shall be made to meet the requirements of 3.15.1.

4.9.21 Mechanical - output.- When the mechanical calibration input shaft is positioned to the specified values by the special test equipment the position of the periscope drive mount output shaft, when indicated shall conform to the values and tolerances of Table VI as specified in 3.15.2.

4.9.22 Electrical - output.- The coarse and fine synchro position shall be indicated by special testing equipment capable of reading angular rotation in degrees to the specified values and tolerances when the mechanical calibration output shaft is positioned in accordance with Table VII and 3.15.2.2.

4.9.23 Backlash.- The backlash shall be inspected on special test equipment. Visual observation of the output shafts shall be made and must meet the angular requirements of 3.16.

4.9.24 Water resistance.- After exposing the computer to the water resistance test as specified in 3.17.3, the computer shall be internally inspected and shall show no evidence of water penetration or damage.

4.9.25 Physical failure, post environmental.- After thermal equilibrium has been reached at the low and high temperatures specified, the computer shall be returned to room temperature (+60°F to +90°F) and visually inspected for any physical failure of metal and glass components and shall meet the requirements of 3.18.

4.9.26 Illumination.- Illumination shall be visually observed in a totally darkened area to meet the requirements of 3.19.

4.9.27 Interrupter switches.- When the computer hand crank, reset button and ammo selector handle are actuated as specified, the results shall be visually observed to meet the requirements of 3.20.

4.9.28 High potential.- This test shall be conducted in accordance with Standard MIL-STD-202 and shall meet the requirements of 3.21.1.

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4.9.28.1 Insulation resistance.- Immediately following the high potential test, this test shall be conducted in accordance with Standard MIL-STD-202 and shall meet the requirements of 3.21.2.

4.9.28.2 Continuity.- This test shall be conducted using standard measuring equipment to inspect each circuit of the wiring diagram for possible open or breaks in wiring as required in 3.21.3.

4.9.28.3 Electrical zero.- The coarse and fine synchros shall be electrically zeroed with respect to each other as specified in the applicable drawings as specified in 3.21.4 and Specification MIL-S-16892.

4.9.29 Identification of components.- A visual inspection of the serial numbers stencilled on the servo amplifier shall be made to insure that it is the component that has been qualified with the computer as required by 3.22.

4.9.30 Identification marking.- A visual inspection shall be made of the method of identification used to insure conformance to 3.22.

4.9.31 Packaging, packing and marking.- Quality assurance provision for packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8602750A. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Ballistics Computer M14 with range servo amplifier is used to receive an electrical signal of the log of range data from a range-finder synchro transmitter thereby producing induced voltage at the synchro transformer in the computer. Signal is relayed and amplified in the range servo amplifier to cause operation of the computer. The computer computes superelevation for various ammunitions and transmits same mechanically and electrically to the other components of the gun sighting system of the 120 MM Gun Tank M103A1.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specifications.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for initial production sample including date of delivery (see 3.1).

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6.3 Definitions.

6.3.1 Special test equipment (STE).- Special testing equipment is defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.3.2 Standard measuring equipment (SME).- Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

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